

# Majid Ghasemi

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## Research interests

Social Reinforcement Learning, Multi-agent Reinforcement Learning, Machine Ethics, AI Alignment, and AI Ethics

## Education

**University of Waterloo** Waterloo, ON, Canada

**Ph.D.** in Electrical & Computer Engineering Sep 2025 – Present

Supervised by Mark Crowley

Research area: *Ethical decision making in multi-agent systems and how social learning can be a way to achieve it*

**Wilfrid Laurier University (WLU)** Waterloo, ON, Canada

**MASc** in Applied Computing Sep 2023 – May 2025

Supervised by Dariush Ebrahimi. *GPA: A+.*

Thesis title: *Reinforcement Learning for Scheduling and Routing: Electric Vehicles Charging and Patrol Scheduling*

**Imam Khomeini Intl. University (IKIU)** Qazvin, Iran

**B.Sc.** in Software Engineering Sep 2017 – March 2022

Supervised by Mahdi Bahaghighat. *GPA: A-.*

## Publications

### Learning When to Trust in Contextual Social Bandits

*Majid Ghasemi, and Mark Crowley.*

*19th European Workshop on Reinforcement Learning*

### Real-Time Electric Vehicle Routing and Charging Optimization with Battery Life Consideration

*Dariush Ebrahimi, Majid Ghasemi, Mohammad Kashefi, and Fadi Alzhouri.*

*Internet of Things journal*

### Rethinking AI Alignment: From Static Rewards to Social Reinforcement Learning

*Majid Ghasemi, and Mark Crowley.*

*ICML 2026 Pluralistic ALignment Workshop*

### Can Standard MARL Metrics Distinguish Communicative from Strategic Action?

*Majid Ghasemi, and Mark Crowley.*

*ICML 2026 PhilML Workshop*

**Citation count prediction based on Google Scholar profiles and Clarivate's journal citation reports**

Mahdi Bahaghighat, Leila Akbari, *Majid Ghasemi*, and Qin Xin.  
*Information Research journal*

**Toward Virtuous Reinforcement Learning: A Critique and Roadmap**

*Majid Ghasemi*, and Mark Crowley.  
*Machine Ethics: from formal methods to emergent machine ethics workshop at the AAAI conference*

**Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response**

*Majid Ghasemi*, and Dariush Ebrahimi.  
*IECON 2025 – 51st Annual Conference of the IEEE Industrial Electronics Society.*

**Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety**

*Majid Ghasemi*, Ebrahim Sarkhouh, Fadi Alzhouri, and Dariush Ebrahimi.  
*2025 IEEE 34th International Symposium on Industrial Electronics (ISIE).*

**Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging**

*Majid Ghasemi*, Fadi Alzhouri, and Dariush Ebrahimi.  
*2025 21st International Conference on the Design of Reliable Communication Networks (DRCN).*

**Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Challenges**

*Majid Ghasemi*, Amirhossein Mousavi, and Dariush Ebrahimi.  
*Preprint on Arxiv.*

**Introduction to Reinforcement Learning**

*Majid Ghasemi*, and Dariush Ebrahimi.  
*Preprint on Arxiv.*

**Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm**

Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, *Majid Ghasemi*, Fadi Alzhouri, and Dariush Ebrahimi.  
*2024 International Conference on the Dynamics of Information Systems.*

**A high-accuracy phishing website detection method based on machine learning**

Mahdi Bahaghighat, *Majid Ghasemi*, and Figen Ozen.

|                            |                                                                                                                                                                                                                                                                                                                                                                                     |                            |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Honors and<br>scholarships | AIES conference Student Program winner                                                                                                                                                                                                                                                                                                                                              | 2026                       |
|                            | GSRDA Award x2 (University of Waterloo)                                                                                                                                                                                                                                                                                                                                             | 2026                       |
|                            | Int'l Doctoral Student Award (University of Waterloo)                                                                                                                                                                                                                                                                                                                               | 2025 – 2029                |
|                            | Winner of the Academic Excellence Gold Medal (WLU)                                                                                                                                                                                                                                                                                                                                  | 2025                       |
|                            | Graduate Scholarship (WLU)                                                                                                                                                                                                                                                                                                                                                          | 2023 – 2025                |
|                            | Graduate Fellowship (WLU)                                                                                                                                                                                                                                                                                                                                                           | 2023 – 2025                |
|                            | Differential Tuition Fee Waiver (WLU)                                                                                                                                                                                                                                                                                                                                               | 2023 – 2025                |
|                            | Excellence Students Tuition Fee Waiver (IKIU)                                                                                                                                                                                                                                                                                                                                       | 2017–2022                  |
| Research experience        | <b>UWECML/CompThink lab</b>                                                                                                                                                                                                                                                                                                                                                         |                            |
|                            | Supervisor: Prof. Mark Crowley (University of Waterloo) Sep 2025 – Present<br>Developing socially informed and ethically grounded approaches to multi-agent reinforcement learning.                                                                                                                                                                                                 |                            |
|                            | <b>Edge Computing lab</b>                                                                                                                                                                                                                                                                                                                                                           |                            |
|                            | Supervisor: Dr. Dariush Ebrahimi (WLU)                                                                                                                                                                                                                                                                                                                                              | Sep 2023 – May 2025        |
|                            | Conducted research on reinforcement learning for intelligent transportation systems, focusing on EV fleet management and routing optimization.                                                                                                                                                                                                                                      |                            |
| Teaching experience        | <b>AIST lab</b>                                                                                                                                                                                                                                                                                                                                                                     |                            |
|                            | Supervisor: Dr. Mahdi Bahaghighat (IKIU)                                                                                                                                                                                                                                                                                                                                            | Sep Jan. 2021 – March 2022 |
|                            | Researched on phishing website detection with machine learning.                                                                                                                                                                                                                                                                                                                     |                            |
|                            | <b>Teaching Assistant, IKIU, WLU, and UW</b>                                                                                                                                                                                                                                                                                                                                        |                            |
|                            |                                                                                                                                                                                                                                                                                                                                                                                     | 2023-Present               |
| Industry experience        | Courses including AI, Data Mining, Algorithm Design and Analysis, and Discrete Mathematics                                                                                                                                                                                                                                                                                          |                            |
|                            | <b>Kisoji Biotechnology Inc.</b> , Machine Learning Department Waterloo, ON, Canada                                                                                                                                                                                                                                                                                                 |                            |
|                            | AI Research Intern                                                                                                                                                                                                                                                                                                                                                                  | May 2026 – Present         |
|                            | Currently building the Expressibility Engine — a 3-module system for antibody expression optimization: domain-adaptive prediction with Bayesian project embeddings, RL-guided experimental design (GRPO + DPP diversity), and sparse graph-based causal driver discovery. Reduces wet-lab rounds via active learning while providing interpretable guidance for antibody engineers. |                            |
|                            | <b>(R) Stands for Reviewer, (PC) stands for Program Committee</b>                                                                                                                                                                                                                                                                                                                   |                            |
| Service to the Community   | (PC) AAAI/ACM Conference on AI, Ethics, and Philosophy (AIES) 2026                                                                                                                                                                                                                                                                                                                  |                            |

(R) Reinforcement Learning in Big Worlds at Reinforcement Learning Conference (RLC) 2026  
(R) PhilML Workshop at ICML 2026  
(R) Pluralistic Alignment Workshop at ICML 2026  
(R) Post-AGI Science and Society (P-AGI) Workshop at ICLR 2026  
(R) IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)  
(R) ACM Transactions on Intelligent Systems and Technology

## Grants

**HU University of Applied Sciences Utrecht** Aug. 2022 – Aug. 2023

Recipient of a grant to conduct research on Woman, Life, Freedom movement in Iran with computational social science and machine learning approaches.